

FlyRank Capstone Demo Video — Word-by-Word Script & Walkthrough

Applied Search Intelligence: Predicting Organic Content Decay (3.08× Precision Lift)

Author: Fatima Javaid | **Track:** Applied Search Intelligence (FlyRank ML Internship)
Deliverable: Assignment 8.1 (FL-09) & 8.2 (FL-10) — 3–5 Min Live Demo Narration Script
Live Repository: https://github.com/Fatima-38/Flyrank_ML_Internship | **Live Paper:** https://fatima-38.github.io/Flyrank_ML_Internship/

1. Pre-Recording Preparation Checklist

Item	What to Open / Setup	Purpose & Action
Window 1	VS Code / Terminal: Path: Flyrank_ML_Internship Command: python scripts/run_all.py	Run live end-to-end pipeline on camera. Demonstrates feature prep, data contract validation & LightGBM training.
Window 2	Live Research Paper: https://fatima-38.github.io/Flyrank_ML_Internship/	Showcase live evaluation benchmark table (Precision@50: 0.740 vs 0.240) and Action Playbook.
Window 3	Generated Outputs: outputs/refresh_queue_sample.csv	Showcase actual triage action queue (ACT_01 to ACT_04) with decay probabilities and reason codes.
Recording Tool	Loom / OBS Studio / Zoom (Screen Share + Webcam Bubble)	Record screen with narration. Strict Rule: NO slides — show live code, terminal, and browser.

2. 5-Minute Video Timeline Breakdown

Timestamp	Segment Title	Screen Display (Visual)	Key Rubric Criteria Met
0:00 – 0:45	Intro & Problem Framing	GitHub Repo & Live Paper Header	Explain what it does, for whom & why heuristic alerts fail (76% false alarms).
0:45 – 2:00	Live Pipeline Execution	Terminal: python scripts/run_all.py	Live execution of data contracts, leakage check, and LightGBM model training.
2:00 – 3:00	Key Design Decision	Live Paper: Benchmark Metrics Table	One Design Decision on camera: GroupShuffleSplit on client domains to stop test leakage.
3:00 – 4:00	Key Limitation & Playbook	refresh_queue_sample.csv & Playbook	One Limitation on camera: Low-impression query variance & automated No-Go rules.
4:00 – 4:45	Conclusion & Deliverables	GitHub Pages & Portfolio links	Transparency diligence, open-source deliverables, and closing summary.

3. Word-by-Word Narration Script

Read or adapt the following script naturally. Keep your delivery confident and clear.

Segment 1: Introduction & Problem Statement (0:00 – 0:45)

[VISUAL: Display GitHub Repository or Live Research Paper Header]

"Hi everyone, I'm Fatima Javaid, a Computer Science undergraduate at UVAS Lahore and a Machine Learning Intern at FlyRank. Today, I'm excited to walk you through my capstone project: **Applied Search Intelligence for Predicting Organic Content Decay**.

In enterprise SEO, websites with tens of thousands of URLs face what we call 'silent organic decay'—high-value pages gradually lose rankings and organic traffic without triggering standard technical health alerts. Traditional monitoring relies on brittle heuristics—such as flagging a page if its position drops by more than 5 spots. However, because Google rankings oscillate naturally, heuristic rules yield a Precision@50 of only 0.24. That means nearly 76% of editorial alerts are false alarms, wasting hundreds of content writing hours.

Our goal was to engineer a machine learning decision-support system on 30,000 enterprise search URLs that accurately predicts decaying assets and tells teams exactly what action to take."

Segment 2: Live Pipeline Execution & Architecture (0:45 – 2:00)

[VISUAL: Switch to Terminal / VS Code and execute python scripts/run_all.py]

"Let's run the end-to-end pipeline live in the terminal using python scripts/run_all.py.

Here is what the pipeline is executing under the hood:

- **Step 1 — Ingestion & Contract:** Ingests 30,000 anonymized search URLs and validates schema types under strict DuckDB data contracts.
- **Step 2 — Leakage Audit:** In Week 3, we audited features and purged target proxies like trend_direction and trend_pct to guarantee zero post-observation data leakage.
- **Step 3 — Feature Engineering:** Computes 44 engineered ranking velocity, CTR delta, and impression volatility signals.
- **Step 4 — Model Training:** Trains our tuned LightGBM classifier with GBDT objective and evaluates under cross-domain holdout splits.

As you can see live in the terminal output, the model has trained in under 5 seconds, validated all constraints, and exported our prediction queues."

Segment 3: Key Design Decision Explained (2:00 – 3:00)

[VISUAL: Switch to Live Paper and highlight Benchmark Metrics Table]

"Now, I want to highlight **one critical technical design decision** we made:

Instead of using a standard random train_test_split, we implemented a **GroupShuffleSplit based on Client Domain IDs**, completely holding out 20% of unseen enterprise accounts from training.

Why? Because URLs belonging to the same client share domain authority, backlink profiles, and technical infrastructure. A naive random split would leak domain-level signals between train and test sets, artificially inflating accuracy.

Under this strict, group-aware validation, our LightGBM model achieved a **Precision@50 of 0.740 compared to 0.240 for**

the heuristic baseline—a verified 3.08× precision lift on completely new client domains."

Segment 4: Key Limitation & Action Playbook (3:00 – 4:00)

[VISUAL: Open outputs/refresh_queue_sample.csv or Action Playbook in live paper]

*"To maintain scientific credibility, I also want to explain **one key limitation of this build** on camera:*

For long-tail URLs with fewer than 100 monthly impressions, CTR calculations suffer from high statistical variance—a single accidental click can cause a 50% swing in CTR that looks like volatility but is just small-sample noise.

*To solve this limitation in production, we engineered automated '**No-Go Guardrails**'. Any URL with insufficient impression volume is filtered out from manual rewrites and assigned to automated technical checks.*

For qualified URLs, the system automatically triages pages into 4 actionable Playbook tiers:

- **ACT_01 (Full Structural Rewrite):** For high-value URLs where search intent has evolved.
- **ACT_02 (Title & Meta CTR Optimization):** For URLs with high impressions but declining click-through rates.
- **ACT_03 (Internal Linking & Schema):** For pages losing rank due to internal link depth.
- **ACT_04 (Prune / 301 Consolidation):** For pages with near-zero traffic over multiple quarters."

Segment 5: Conclusion & Live Proof (4:00 – 4:45)

[VISUAL: Show GitHub Pages Live Site & Author Portfolio at fatimajavaid.vercel.app]

"Everything you see here—the full research paper, reproducible notebooks from Week 1 to Week 8, and automated scripts—is open-source on GitHub at [Fatima-38/Flyrank_ML_Internship](#) and live on GitHub Pages.

This project transformed how I approach machine learning: prioritizing decision cost and top-k precision over abstract accuracy, and building systems that directly serve human editorial workflows.

Thank you to FlyRank for the dataset and mentorship throughout this journey. Thank you for watching!"

4. Capstone Checkpoint (FL-09 / FL-10) Evaluation Rubric Checklist

- **Reproducibility:** Stranger can clone repo, run python scripts/run_all.py, and verify output without errors.
- **No Slides Rule:** Video strictly recorded showing live code execution, data files, and interactive browser reports.
- **Design Decision on Camera:** Explicitly articulated why GroupShuffleSplit prevents domain leakage.
- **Limitation on Camera:** Explicitly articulated low-impression CTR variance and No-Go guardrails.
- **Transparency Diligence:** Named AI pair-programming scope vs independent student engineering.
- **Public Proof:** Live site on FlyRank GitHub Pages + Build-in-Public LinkedIn story.